Atomic radius (Å)	1.52	1.80	1.90	2.06	1.97
Density ($g.cm^{-3}$ at 293 K)	1.3×10^{-3}	2.07	4.81	6.23	9.20
Melting point (K)	54	388	494	723	527
Boiling point (K)	90	718	958	1261	1235

3.2 Oxygen:

Preparation: The atmosphere and water contain 23% and 83% by mass of oxygen respectively. Most of the world's rock contain combined oxygen. Industrially oxygen is obtained by fractional distillation of liquefied air. In the laboratory, oxygen is prepared by one of the following methods.

The decomposition of hydrogen peroxide in the presence of catalyst (MnO_2) or by oxidation with potassium permanganate.

$$2H_2O_2 \rightleftharpoons 2H_2O + O_2$$

$$5H_2O_2 + 2MnO_4^- + 6H^+ \longrightarrow 5O_2 + 8H_2O + 2Mn^{2+}$$

The thermal decomposition of certain metallic oxides or oxoanions gives oxygen.

$$2HgO \xrightarrow{\Delta} 2Hg + O_2$$

$$2BaO_2 \xrightarrow{\Delta} 2BaO + O_2$$

$$2KClO_3 \xrightarrow[MnO_2]{\Delta} 2KCl + 3O_2$$

$$2KNO_3 \xrightarrow{\Delta} 2KNO_2 + O_2$$

Properties

Under ordinary condition oxygen exists as a diatomic gas. Oxygen is paramagnetic. Like nitrogen and fluorine, oxygen form strong hydrogen bonds. Oxygen exists in two allotropic forms namely dioxygen (O_2) and ozone or trioxygen (O_3). Although negligible amounts of ozone occurs at sea level it is formed in the upper atmosphere by the action of ultraviolet light. In the laboratory ozone is prepared by passing electrical discharge through oxygen. At a potential of 20,000 V about 10% of oxygen is converted into ozone it gives a mixture known as ozonised oxygen. Pure ozone is obtained as a pale blue gas by the fractional distillation of liquefied ozonised oxygen.

$$\underset{\text{Oxygen}}{O_2} \rightleftharpoons \underset{\text{atomic oxygen}}{2(O)}$$

$$O_2 + (O) \rightleftharpoons \underset{\text{Ozone}}{O_3}$$

The ozone molecule has a bent shape and symmetrical with delocalised bonding between the oxygen atoms.

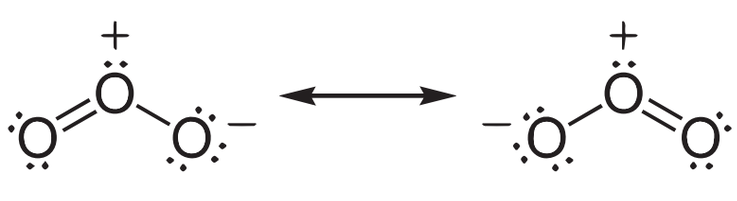


Figure 3.8 Structure of ozone

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